

Cambridge Underground Token Factory

The AI data center of 2035 is not in orbit. It is 50 m under our feet.

Track 1 · SundAI × MIT AI Hackathon · October 2026 . Kalyani Guggilla

Power, not money, is the hard limit

$30B

five-year budget for chips, buildings, energy and launches

500 MW

of Earth grid power across all sites together

6 sites

maximum on Earth; orbit costs $200 per kg to launch

4.45B

users across three cases: 4B prompters, 50M agent developers, 400M knowledge workers

Scored on users served within the limits, then on dollars saved. Every watt lost to cooling or conversion is a user we cannot serve.

Sources: SundAI × MIT AI Hackathon challenge brief (Oct 2026); user counts and limits as set by the organizers.

Three ideas, one underground plant

Cool with river and aquifer water

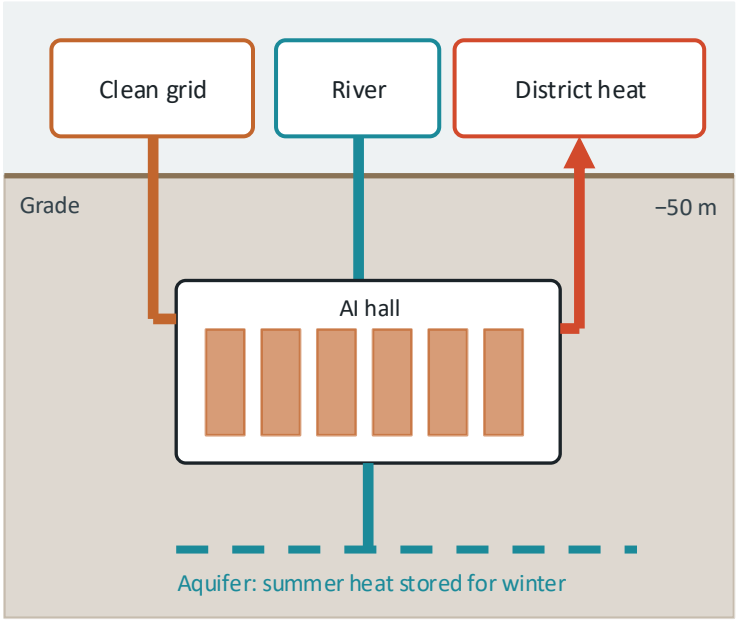
Liquid cooling dumps heat into Charles River water and an aquifer cold well, and the bedrock stores summer heat for winter. No cooling towers, no PFAS fluids.

Sell the heat

Heat pumps feed Cambridge district heating: about 31 MW of heat per 100 MW of IT, roughly 25,000 homes.

Clean power, matched hourly

Offshore wind, storage and firm clean power at 95% hourly, delivered on an 800 V DC bus (97% efficient).



Backed by research: liquid cooling gives 50–60 °C reusable heat [1]; heat pumps lift it for district heating [2]; aquifer storage cools ~7× more efficiently than chillers at summer peak [3]; Meta Odense already heats 12,000+ homes this way [15].

Every piece already runs somewhere

Iron Mountain, Pennsylvania

Data halls about 22 stories underground in a former limestone mine, cooled by an underground lake at ~50 °F. [16]

Lefdal Mine, Norway

Data center inside a former mine, cooled with fjord water and run on hydropower. [17]

Meta Odense, Denmark

About 45 MW of heat pumps lift waste heat to 70–75 °C; heats 12,000+ homes. [15]

Stockholm and Espoo

Stockholm Data Parks heats ~2,500 apartments [1]; Microsoft and Fortum plan ~40% of Espoo-area district heat. [18]

What is new in ours: all of these together, at 100 MW, next to a city that buys the heat.

Sources: see the References slide for full citations.

MIT already runs on district heat

100+

campus buildings get steam heat, electricity and chilled water from one plant [21]

2 × 22 MW

gas turbines with heat-recovery boilers, upgraded 2019–2021 [20][21]

2050

target for no direct campus emissions; net zero by 2026 [22][23]

What MIT is studying to decarbonize

- Heat pumps and low-temperature district heating
- Geothermal boreholes as a thermal battery
- River-water heat exchange and energy storage
- Waste-heat recovery and “green steam” partners [23][24]

Next door: the Kendall plant

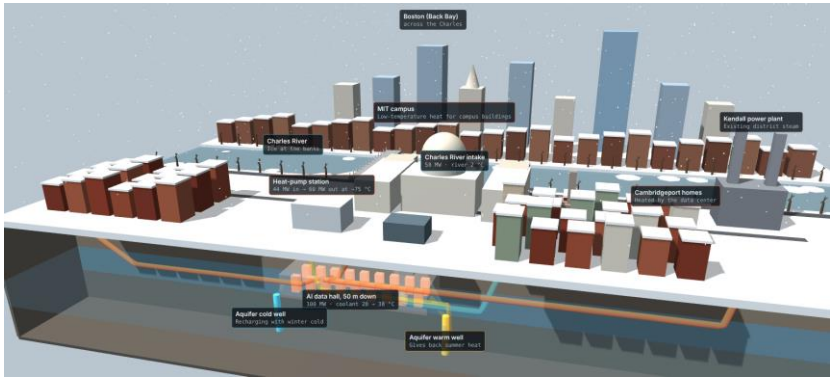
- 250 MW cogeneration plant serving Cambridge and Boston [25]
- 42 MW electric boiler added in 2024 [25]
- 35 MW Charles River heat pump under construction, through 2028 [26]

Where we fit: our 38 °C server heat plus heat pumps is the clean heat both networks are looking for.

Sources: MIT Facilities [20]; MIT News [21]; IDEA [22]; MIT Sustainability [23]; Inside Climate News [24]; ENR [25]; pv magazine [26].

One plant, two seasons

Winter · January

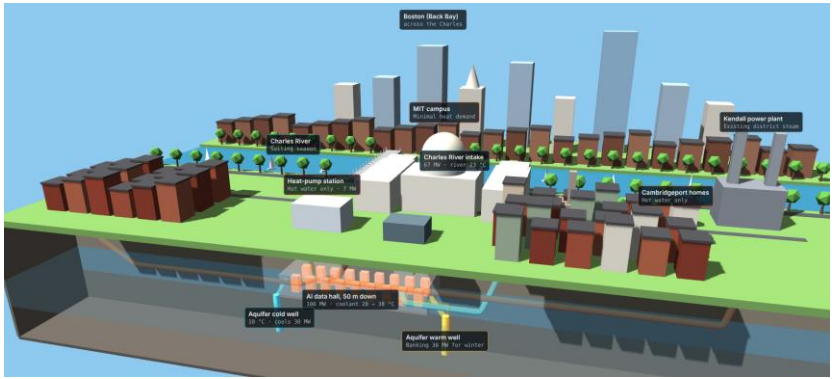


60 MW heat to homes and MIT

58 MW to the Charles River

Gives back summer heat from the aquifer

Summer · July



7 MW hot water only

67 MW to the Charles River

30 MW banked in the aquifer

Live 3D model: drag to orbit, switch seasons, watch the heat flow to homes.

Sources: monthly flows for one 100 MW plant from our twin; aquifer storage [3][5]; river heat pumps [26]. Stylized, not to scale.

How we cool it: liquid in, heat out, no water burned

1 Cold plates on every chip

Warm water (~35 °C) flows over each GPU and leaves ~10 °C hotter. No fans, no chillers, no cooling towers.

2 Sealed loop to a heat exchanger

Chip water never leaves the loop. A plate exchanger hands the heat to the outside sinks.

PUE 1.09

all year, even in July

0.009 mL

water per prompt (vs 0.49 mL)

3 Where the heat goes (per 100 MW plant)

WINTER · JAN

60 MW

to homes + MIT via heat pumps at 75 °C

58 MW

to the Charles River

SUMMER · JUL

30 MW

into a 10 °C aquifer cold well, saved for winter

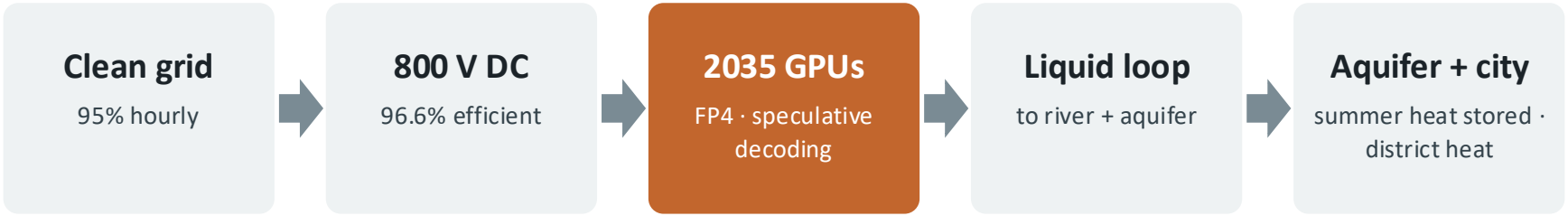
67 MW

to the Charles River

Hard limits in the twin: river ≤ 100 MW, aquifer 30 MW, boreholes ~10 MW. Dry coolers take any overflow.

Sources: heat reuse [1]; heat pumps [2]; aquifer storage [3][5]; cooling life cycle [4]; river heat pump [26]. MW from our twin.

A digital twin of the whole plant



8,760

hours of Cambridge weather and bedrock simulated

1,696

feasible designs searched by the optimizer

2

live demos: fleet planner + plant twin, chiplet thermal

Sources: organizers' chip figures; usage [9][10][11]; heat physics [1][2][3]. Results are model outputs.

Versus a 2025-style data center

4.3×

less energy per token

0.30 → 0.07 J per output token in the plant twin

1.09

PUE, down from ~1.28

stays at 1.09 even in July; dry coolers hit 1.11

50×

less water

0.49 → 0.009 mL per prompt; nothing evaporated on site

7×

less CO_2

136 → 20 kt a year per 100 MW at 95% hourly clean power

Where the 4.3× comes from: the serving stack and 800 V DC give 3.5× in any building. The site (liquid cooling to river and aquifer, cooler chips, heat sales) cuts the rest by a further 20%.

Sources: our twin. For context, liquid cooling cuts life-cycle emissions 15–21% and water 31–52% vs air [4]; AI water footprint [6].

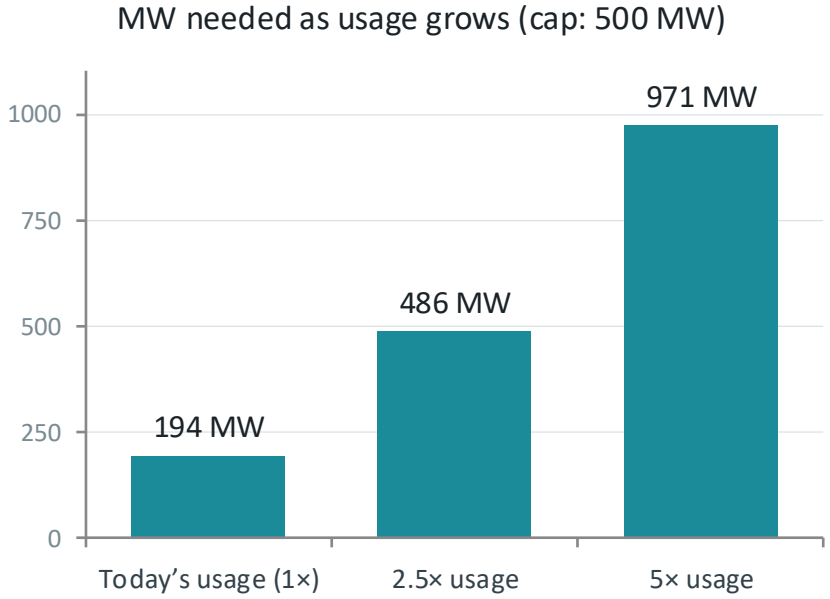
All 4.45B users, under a quarter of the budget

4.45B users served: all three cases at 100%

$6.8B of $30B over 5 years

194 MW of 500 MW at summer peak, on 2 sites

2.6× usage growth before the cap binds



Above 500 MW: drop agent developers first, or overflow to orbit

Sources: usage per user from OpenAI/NBER [9], Claude Code docs [10], St. Louis Fed [11], reply length [8]; token volume context [12].

Why Cambridge beats orbit for the core fleet

	Cambridge underground	Low Earth orbit
Cooling	River + aquifer water; PUE 1.09 even in July	Radiators only; large and heavy
Energy	Clean grid at ~$106/MWh	Free sunlight
Up-front cost	~$13M per MW built	Launch $200/kg + space hardware
Repairs	Technicians on site	None; failed chips stay failed
Latency	A few ms to US users	Higher, varies with orbit
Waste heat	Sold to 25,000 homes per 100 MW	Radiated into space

Orbit stays in the plan as overflow if 2035 usage grows past 2.6× today.

Sources: launch cost from the challenge brief; heat reuse in practice [15][18]; underground operation [16][17].

The best AI factory is under our feet

4.45B users · 194 MW · $6.8B · 4.3× less energy per token · 50× less water

Live demo and code on GitHub · every number editable, every source cited



Research papers & citations

Cambridge Underground Token Factory

Word document · 26 sources + 5 background readings

References

[1] Yuan et al. 2023, Renewable & Sustainable Energy Reviews (journal)
[2] Wang et al. 2024, Energy and Buildings (ORNL) (journal)
[3] Oh, Sickinger & Acero-Allard 2025, Applied Energy (NREL) (journal)
[4] Microsoft 2025, Nature (journal)
[5] Applied Energy 2026 (PolyU & DTU), ATES + heat recovery (journal)
[6] Li et al., Making AI Less "Thirsty" (arXiv) (preprint)
[7] Google 2025, Life-cycle emissions of AI hardware (arXiv) (preprint)
[8] LLM response lengths study 2025 (arXiv) (preprint)
[9] OpenAI & NBER 2025, How People Use ChatGPT (report)
[10] Anthropic, Claude Code docs: Manage costs (report)
[11] St. Louis Fed 2025, Generative AI and work productivity (report)
[12] Google I/O 2026 keynote (report)
[13] Princeton ZERO Lab, 24/7 clean energy (via Canary Media) (report)
[14] Duke 2025, Rethinking Load Growth (via POWER) (report)
[15] Ramboll: Meta Odense heat to district heating (industry)
[16] Network World: Iron Mountain underground data center (industry)
[17] DCD: Lefdal Mine Datacenter, Norway (industry)
[18] Eurelectric: Microsoft & Fortum, Espoo (industry)
[19] HPCwire 2023: 3M PFAS exit and two-phase cooling (industry)
[20] MIT Facilities: Central Utilities Plant (report)
[21] MIT News 2017: Central Utilities Plant upgrade (report)
[22] IDEA 2022: How MIT is cutting carbon on campus (industry)
[23] MIT Sustainability 2023: district energy decarbonization (report)
[24] Inside Climate News 2025: MIT thermal energy options (industry)
[25] ENR 2024: Kendall plant 42 MW electric boiler (industry)
[26] pv magazine 2025: Charles River heat pump complex (industry)

Journal = peer-reviewed. Full citations with links are in the playbook and README.


THANK YOU!